A

Real Time Project Report

on

"HOUSE WITH EARTH QUAKE WARNING ALARM"

Submitted in partial fulfilment of the requirement for the award of degree

of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

submitted by

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Under the Guidance of

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SWARNA BHARATHI INSTITUTE OF SCIENCE & TECHNOLOGY

(Accredited by NAAC-A+ Approved by AICTE, Affiliated to JNTUH, Hyderabad)

Pakabanda Bazar, Mamtha Hospital Road, Khammam-507002

(2025-26)

SWARNA BHARATHI INSTITUTE OF SCIENCE & TECHNOLOGY, KHAMMAM

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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

(2025-26)

CERTIFICATE

This is to certify that the Real Time Project entitled **"House with Earth Quake Warning Alarm"** *is a bonafide record of work carried out by*

V. PAWAN KALYAN (25M65A0409)

We hereby accord our approval of it as a Real Time Project carried out and presented in a manner required for its acceptance in partial fulfilment for the award of degree of ***BACHELOR OF TECHNOLOGY*** *in* ***ELECTRONICS AND COMMUNICATION ENGINEERING*** of **Jawaharlal Nehru Technological University Hyderabad, Hyderabad** *during the academic year 2025-26.*

UNDER THE GUIDANCE OF	**HEAD OF THE DEPARTMENT**
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Last but not least I thank all the members of Teaching and Non-Teaching Staff, Department of ECE, and all those who have helped me directly or indirectly in completing this *Real Time* project successfully.

DECLARATION

I hereby declare that the *Real Time Project* entitled **"House with Earth Quake Warning Alarm"** recorded in this project is based on my work carried out at the **"SWARNA BHARATHI INSTITUTE OF SCIENCE &TECHNOLOGY", Khammam** during the B.Tech. course.

DATE:

PLACE: Khammam

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ABSTRACT

Earthquake is an inevitable phenomenon that happens quickly and cannot be prevented, but can be warned. An accelerometer ADXL335 sensor is used to detect pre-earthquake vibrations. When a vibration occurs, the accelerometer detects and converts the equivalent ADC value. These digital values are then read by the microcontroller Arduino. Arduino then compares these values with the current values. If the Arduino detects that the standard value is greater than the threshold, it will beep and display a message on the 16x2 LCD indicating the alarm status and the LED will light up at the same time. notification messages are display to the relevant using lcd the Buzzer module. Both sensors are used for fire detection and gas leaks in the building can be detected using a combination of flame and smoke

Key Words*:* Accelerometer, Arduino, Buzzer module, An accelerometer (ADXL335), Analog To Digital converter(ADC) ,Liquid Crystal Display(16x2 LCD), Light emitting Ddiode(LED)

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ACRONYMS

ADC	Analog To Digital converter
AVR	Alf and Vegard's RISC Processor
CNC	Computer Numerical Computer
EEPROM	Electrically Erasable Programmable Read Only Memory
GSM	Global System for Mobile Communication
HWB	Hardware Bootloader
IDE	Integrated Development Environment
IDII	Interaction Design Institute Ivrea (IDII)
LCD	Liquid Crystal Display
LED	Light emitting Diode
PCB	Printed Circuit Board
SMS	Short Message Service
TTL	Transistor–Transistor Logic
USB	Universal Serial Bus

CHAPTER 1

INTRODUCTION

1.1 Objectives of project

Earthquakes are phenomena caused by the movement of tectonic plates, resulting in the thermal motion of the Earth's core colliding with another Earth plate and breaking it [2]. Earthquakes cause loss of life and property. Earthquakes are unexpected events that cannot be prevented, but can be prevented. Pre-earthquake detection using the highly sensitive ADXL335 accelerometer. It is very sensitive to shock and vibration on all three axes. Whenever it detects any shakes or vibrations, it will start the buzzer that vibrations and convert the m into equivalent ADC value. These values are then read by the Arduino and displayed on the 16x2 LCD. The accelerometer is first calibrated by taking environmental vibration samples when the Arduino is turned on. This sample must be subtracted from the actual reading to get the actual reading, so it doesn't show a warning about the vibrations around it. After finding the correct reading, the Arduino compares these values with the predefined maximum and minimum values. If the Arduino detects a change in the preset value of an axis in both directions (negative and positive), the Arduino will emit a sound and display an alarm on the 16x2 LCD and the LED will light up at the same time [4]. You can adjust the sensitivity of the earthquake detector by changing the values given in the Arduino code. SMS notifications are sent to the corresponding mobile number [12] using the GSM module. And two more Using sensors to detect fire, any smoke in the building can be detected using a combination of flame and smoke. If detected, an SMS notification will be sent to the corresponding mobile number [5].

1.2 LITERATURE REVIEW:

In recent years, computing systems have shown a rapid growth, with their explosive evolution benefiting almost all sectors of human activity. Nowadays, mobility has become a highly prized quality. As "smartphone"-type mobile phones are becoming more and more affordable in terms of price, quality and performance, applications developed on the Android platform have the potential to become extremely useful in conjunction with database management systems [11]. The internet of things (IoT) is a concept and, implicitly, a model built on the assumption of the omnipresence of a multitude of objects which, through wireless or cabled connections and unique addressing schemes, can interact and collaborate with other objects, creating new services and applications aimed at reaching a common goal [15]. In this light, the promise of an intelligent environment is extremely seductive, envisioning a world where reality and the artificial digital virtual components of the world interact in order to create a better environment. Being a new Internet revolution, IoT’s goal is to allow things and objects to be connected anytime and anywhere with anyone, using any network, path and service. Because devices can provide information about themselves, objects are recognized and learned by making contextual decisions. They can access information that has been used in a connection with other things, or they can be parts of complex services. The future trend in intelligent packaging is a global expansion in food distribution systems. The quality and safety of vacuum foods are strongly influenced by temperature, making the monitoring of time- temperature history throughout the entire distribution chain compulsory. Commercially available monitoring devices include temperature data loggers, time-temperature integrators (TTIs), smart radio frequency identification (smart-RFID), and others.

CHAPTER 2

IMPLEMENTATION OF PROJECT

2.1 Block diagram:

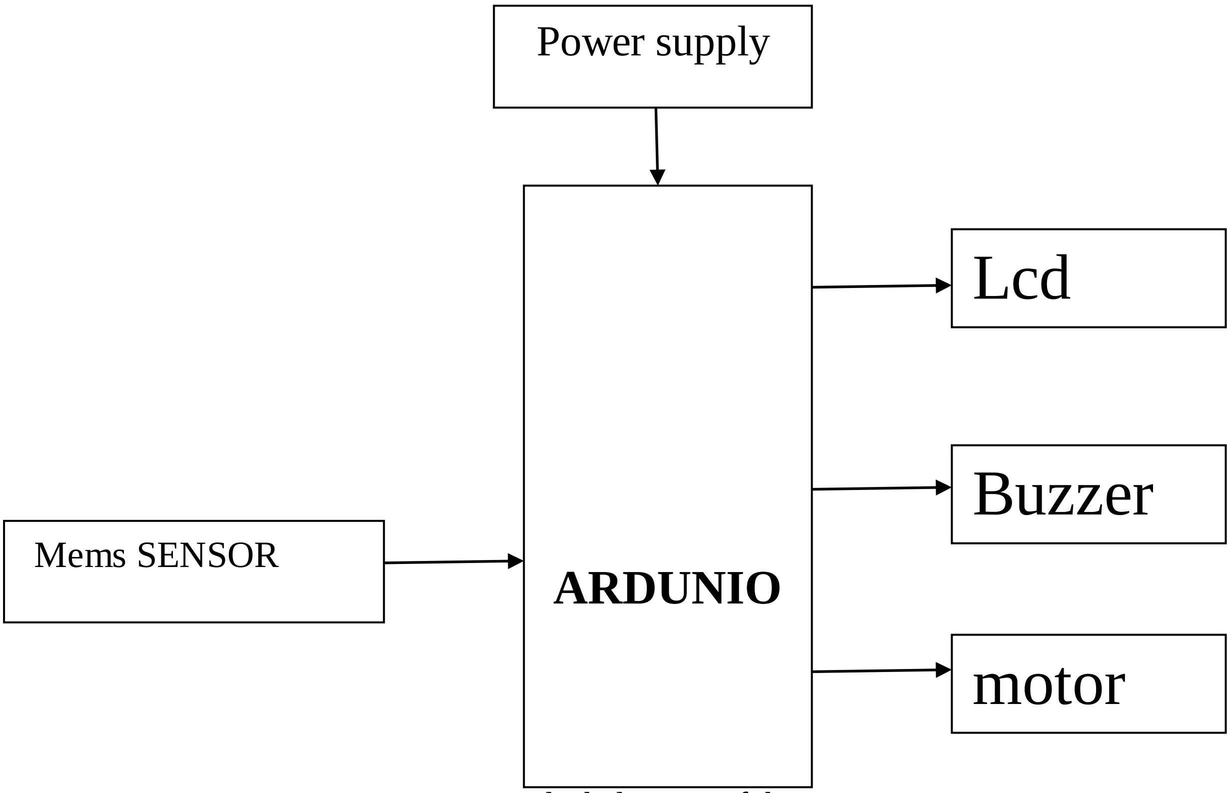

Figure 2.1: Block diagram of the project

2.2 Description of the block diagram:

The ARDUNIO ships as a bare circuit board with keypad, Lcd, and Gsm power supply.

2.3 AVR MICROCONTROLLER

2.31 ARDUINO:

Introduction:

Arduino is a computer hardware and software company, project, and user community that designs and manufactures microcontroller kits for building digital devices and interactive objects that can sense and control objects in the physical world. The project's products are distributed as open-source hardware and software, which are licensed under the GNU Lesser General Public License (LGPL) or the GNU General Public License (GPL), permitting the manufacture of Arduino boards and software distribution by anyone. Arduino boards are available commercially in preassembled form, or as do-it-yourself kits.

Arduino board designs use a variety of microprocessors and controllers. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (*shields*) and other circuits. The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs from personal computers. The microcontrollers are typically programmed using a dialect of features from the programming languages C and C++. In addition to using traditional compiler toolchains, the Arduino project provides an integrated development environment (IDE) based on the Processing language project.

The Arduino project started in 2005 as a program for students at the Interaction Design Institute Ivrea in Ivrea, Italy,[2] aiming to provide a low-cost and easy way for novices and professionals to create devices that interact with their environment using sensors and actuators. Common examples of such devices intended for beginner hobbyists include simple robots, thermostats, and motion detectors.

The name *Arduino* comes from a bar in Ivrea, Italy, where some of the founders of the project used to meet. The bar was named after Arduin of Ivrea, who was the margrave of the March of Ivrea and King of Italy from 1002 to 1014.

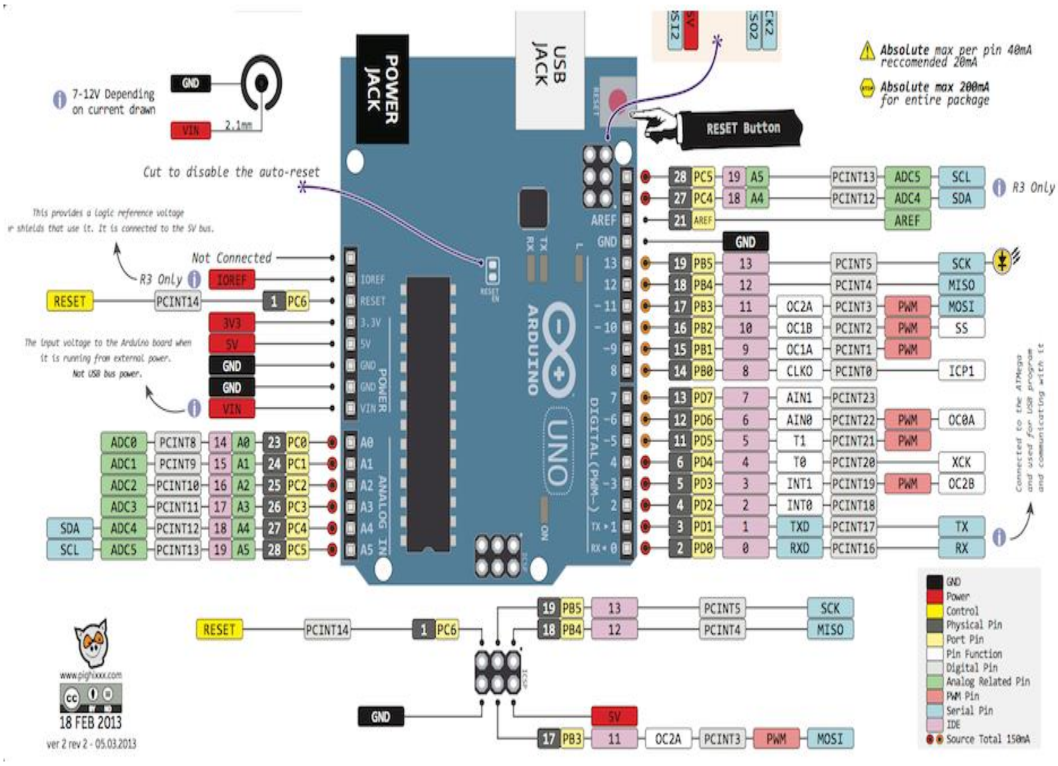

Figure 2.2 Arduino port Diagram

History:

The origin of the Arduino project started at the Interaction Design Institute Ivrea (IDII) in Ivrea, Italy.[2] At that time, the students used a BASIC Stamp microcontroller at a cost of $100, a considerable expense for many students. In 2004, Colombian student Hernando Barragán created the development platform *Wiring* as a Master's thesis project at IDII, under the supervision of Massimo Banzi and Casey Reas, who are known for work on the Processing language. The project goal was to create simple, low-cost tools for creating digital projects by non-engineers. The Wiring platform consisted of a printed circuit board (PCB) with an ATmega168 microcontroller, an IDE based on Processing and library functions to easily program the microcontroller.

In 2005, Massimo Banzi, with David Mellis, another IDII student, and David Cuartielles, added support for the cheaper ATmega8 microcontroller to Wiring. But instead of continuing the work on Wiring, they copied the Wiring source code and renamed it as a separate project, called Arduino.

The initial Arduino core team consisted of Massimo Banzi, David Cuartielles, Tom Igoe, Gianluca Martino, and David Mellis, but Barragán was not invited to participate.

Following the completion of the Wiring platform, lighter and less-expensive versions were distributed in the open-source community.

Adafruit Industries, a New York City supplier of Arduino boards, parts, and assemblies, estimated in mid-2011 that over 300,000 official Arduinos had been commercially produced, and in 2013 that 700,000 official boards were in users' hands.

2.32 Hardware:

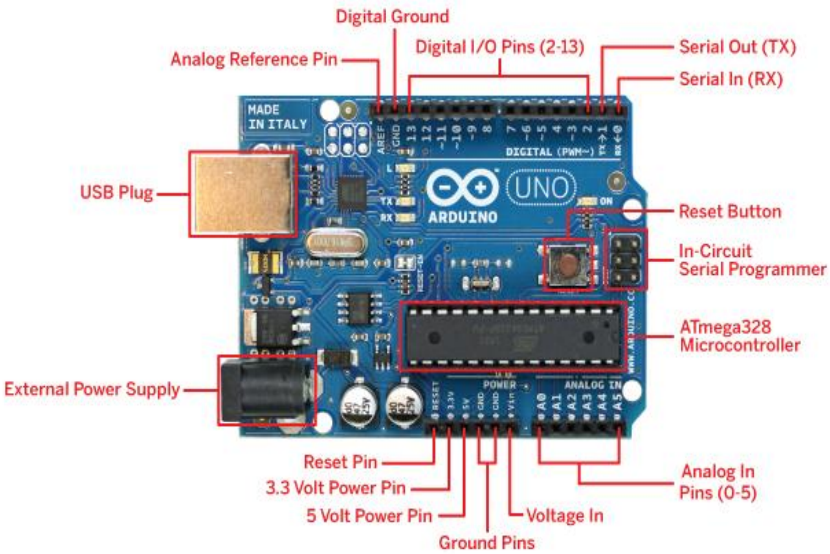

Figure 2.3 Over view of Arduino hardware

Arduino is open-source hardware. The hardware reference designs are distributed under a Creative Commons Attribution Share-Alike 2.5 license and are available on the Arduino website. Layout and production files for some versions of the hardware are also available. The source code for the IDE is released under the GNU General Public License, version 2. Nevertheless, an official Bill of Materials of Arduino boards has never been released by Arduino staff.

Although the hardware and software designs are freely available under copyleft licenses, the developers have requested that the name *Arduino* be exclusive to the official product and not be used for derived works without permission. The official policy document on use of the Arduino name emphasizes that the project is open to incorporating work by others into the official product.[9] Several Arduino-compatible products commercially released have avoided the project name by using various names ending in *-duino*.

An Arduino board consists of an Atmel 8-, 16- or 32-bit AVR microcontroller (ATmega8, ATmega168, ATmega328, ATmega1280, ATmega2560), but other makers' microcontrollers have been used since 2015. The boards use single-row pins or female headers that facilitate connections for programming and incorporation into other circuits. These may connect with add-on modules termed *shields*. Multiple, and possibly stacked shields may be individually addressable via an I²C serial bus. Most boards include a 5 V linear regulator and a 16 MHz crystal oscillator or ceramic resonator. Some designs, such as the LilyPad, run at 8 MHz and dispense with the onboard voltage regulator due to specific form-factor restrictions.

Arduino microcontrollers are pre-programmed with a boot loader that simplifies uploading of programs to the on-chip flash memory. The default bootloader of the Aduino UNO is the optiboot boot loader. Boards are loaded with program code via a serial connection to another computer. Some serial Arduino boards contain a level shifter circuit to convert between RS-232 logic levels and level signals. Current Arduino boards are programmed via Universal Serial Bus (USB), implemented using USB-to-serial adapter chips such as the FTDI FT232. Some boards, such as later-model Uno boards, substitute the FTDI chip with a separate AVR chip containing USB-to-serial firmware, which is reprogrammable via its own ICSP header. Other variants, such as the Arduino Mini and the unofficial Boarduino, use a detachable USB-to-serial adapter board or cable, Bluetooth or other methods, when used with traditional microcontroller tools instead of the Arduino IDE, standard AVR in-system programming (ISP) programming is used.

The Arduino board exposes most of the microcontroller's I/O pins for use by other circuits. The *Diecimila*, *Duemilanove*, and current *Uno*] provide 14 digital I/O pins, six of which can produce pulse-width modulated signals, and six analog inputs, which can also be used as six digital I/O pins. These pins are on the top of the board, via female 0.1-inch (2.54 mm) headers. Several plug-in application shields are also commercially available. The Arduino Nano, and Arduino-compatible Bare Bones Board[13] and Boarduino boards may provide male header pins on the underside of the board that can plug into solderless breadboards.

Many Arduino-compatible and Arduino-derived boards exist. Some are functionally equivalent to an Arduino and can be used interchangeably. Many enhance the basic Arduino by adding output drivers, often for use in school-level education, to simplify making buggies and small robots. Others are electrically equivalent but change the form factor, sometimes retaining compatibility with shields, sometimes not. Some variants use different processors, of varying compatibility.

Official boards

The original Arduino hardware was produced by the Italian company Smart Projects.[15] Some Arduino-branded boards have been designed by the American companies SparkFun Electronics and Adafruit Industries. As of 2016, 17 versions of the Arduino hardware have been commercially produced.

Software development

A program for Arduino may be written in any programming language for a compiler that produces binary machine code for the target processor. Atmel provides a development environment for their microcontrollers, AVR Studio and the newer Atmel Studio.

The Arduino project provides the Arduino integrated development environment (IDE), which is a cross-platform application written in the programming language Java. It originated from the IDE for the languages *Processing* and *Wiring*. It includes a code editor with features such as text cutting and pasting, searching and replacing text, automatic indenting, brace matching, and syntax highlighting, and provides simple *one-click* mechanisms to compile and upload programs to an Arduino board. It also contains a message area, a text console, a toolbar with buttons for common functions and a hierarchy of operation menus.

A program written with the IDE for Arduino is called a *sketch*.[40] Sketches are saved on the development computer as text files with the file extension *.ino*. Arduino Software (IDE) pre-1.0 saved sketches with the extension *.pde*.

The Arduino IDE supports the languages C and C++ using special rules of code structuring. The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures. User-written code only requires two basic functions, for starting the sketch and the main program loop, that are compiled and linked with a program stub *main()* into an executable cyclic executive program with the GNU toolchain, also included with the IDE distribution. The Arduino IDE employs the program *avrdude* to convert the executable code into a text file in hexadecimal encoding that is loaded into the Arduino board by a loader program in the board's firmware.

Applications

- Xoscillo, an open-source oscilloscope[48]
- Arduinome, a MIDI controller device that mimics the Monome
- OBDuino, a trip computer that uses the on-board diagnostics interface found in most modern cars
- Ardupilot, drone software and hardware
- Gameduino, an Arduino shield to create retro 2D video games[49]
- Arduino Phone, a do-it-yourself cell phone[50][51]
- Water quality testing platform
- Automatic titration system based on Arduino and stepper motor[53]
- Low-cost data glove for virtual reality applications[54]
- Impedance sensor system to detect bovine milk adulteration[55]
- Homemade CNC using Arduino and DC motors with close loop control by Homofaciens
- DC motor control using Arduino and H-Bridge

Technical specs

Microcontroller	ATmega328P
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limit)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
PWM Digital I/O Pins	6
Analog Input Pins	6
DC Current per I/O Pin	20 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (ATmega328P) of which 0.5 KB used by bootloader
SRAM	2 KB (ATmega328P)
EEPROM	1 KB (ATmega328P)
Clock Speed	16 MHz
LED_BUILTIN	13
Length	68.6 mm
Width	53.4 mm
Weight	25 g

Figure 2.4 Technical specs

Programming

The Arduino/Genuino Uno can be programmed with the (Arduino Software (IDE)). Select "Arduino/Genuino Uno from the Tools > Board menu (according to the microcontroller on your board). For details, see the reference and tutorials.

The ATmega328 on the Arduino/Genuino Uno comes preprogrammed with a bootloader that allows you to upload new code to it without the use of an external hardware programmer. It communicates using the original STK500 protocol (reference, C header files).

You can also bypass the bootloader and program the microcontroller through the ICSP (In-Circuit Serial Programming) header using Arduino ISP or similar; see these instructions for details.

The ATmega16U2 (or 8U2 in the rev1 and rev2 boards) firmware source code is available in the Arduino repository. The ATmega16U2/8U2 is loaded with a DFU bootloader, which can be activated by:

- On Rev1 boards: connecting the solder jumper on the back of the board (near the map of Italy) and then reseing the 8U2.
- On Rev2 or later boards: there is a resistor that pulling the 8U2/16U2 HWB line to ground, making it easier to put into DFU mode.

You can then use Atmel's FLIP software (Windows) or the DFU programmer (Mac OS X and Linux) to load a new firmware. Or you can use the ISP header with an external programmer (overwriting the DFU bootloader). See this user-contributed tutorial for more information.

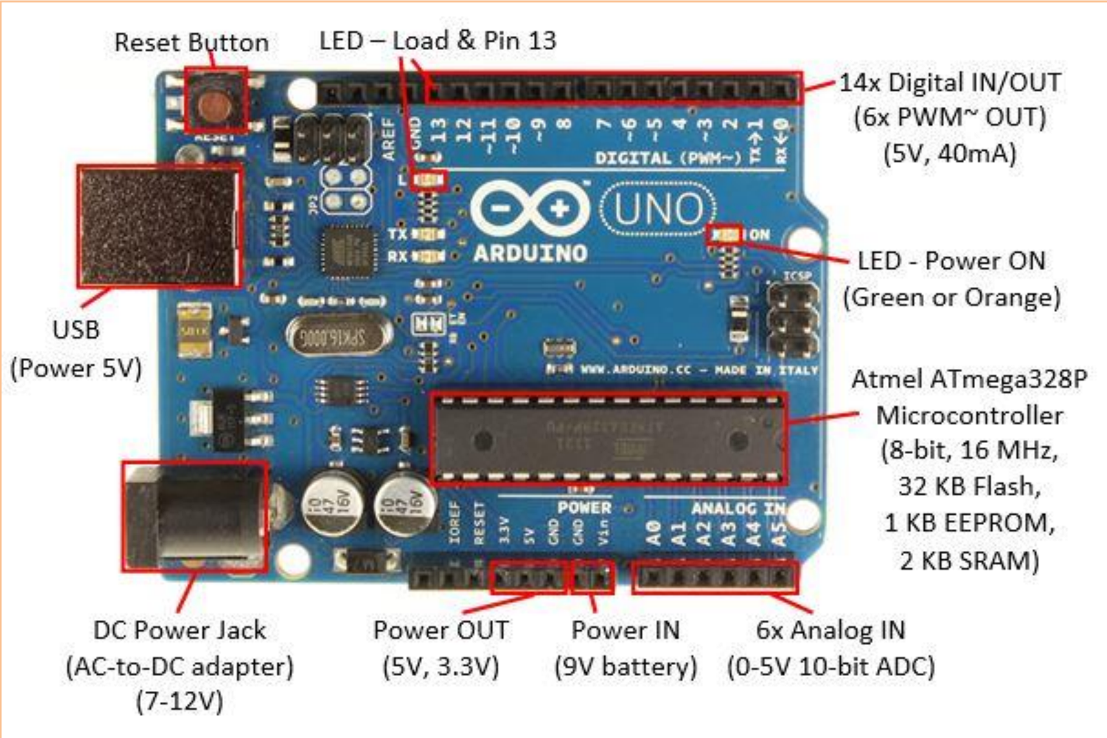

Figure 2.5 Arduino Microcontroller chip

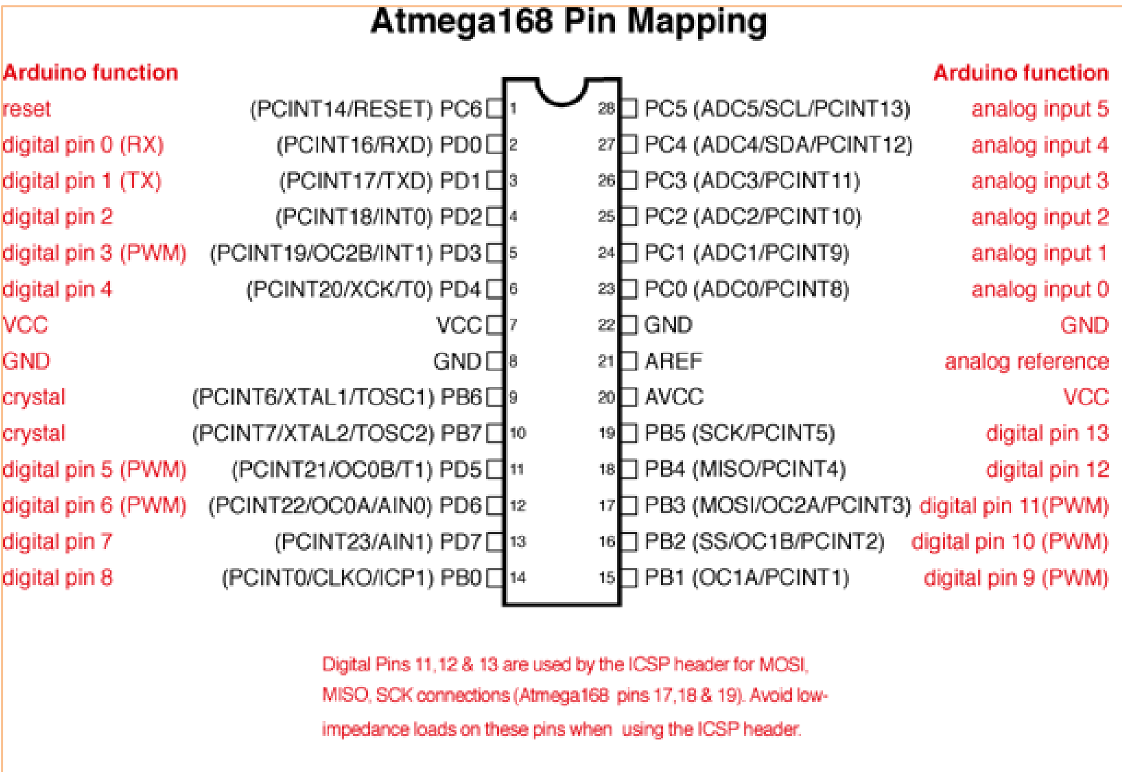

Figure 2.6 Arduino Functional Diagram

The Arduino/Genuino Uno has a resettable poly fuse that protects your computer's USB ports from shorts and overcurrent. Although most computers provide their own internal protection, the fuse provides an extra layer of protection. If more than 500 mA is applied to the USB port, the fuse will automatically break the connection until the short or overload is removed.

Differences with other boards

The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter.

Power

The Arduino/Genuino Uno board can be powered via the USB connection or with an external power supply. The power source is selected automatically.

External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. The adapter can be connected by plugging a 2.1mm centre-positive plug into the board's power jack. Leads from a battery can be inserted in the GND and Vin pin headers of the POWER connector.040 48576663

The board can operate on an external supply from 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may become unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

The power pins are as follows:

Vin. The input voltage to the Arduino/Genuino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.

- 5V. This, pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 - 12V), the USB connector (5V), or the VIN pin of the board (7-12V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator, and can damage your board. We don't advise it.
- 3V3. A 3.3volt supply generated by the on-board regulator. Maximum current draw is 50 mA.

- GND. Ground pins.
- IOREF. This pin on the Arduino/Genuino board provides the voltage reference with which the microcontroller operates. A properly configured shield can read the IOREF pin voltage and select the appropriate power source or enable voltage translators on the outputs to work with the 5V or 3.3V.

Memory

The ATmega328 has 32 KB (with 0.5 KB occupied by the bootloader). It also has 2 KB of SRAM and 1 KB of EEPROM (which can be read and written with the EEPROM library).

Input and Output

See the mapping between Arduino pins and ATmega328P ports. The mapping for the Atmega8, 168, and 328 is identical.

Each of the 14 digital pins on the Uno can be used as an input or output, using pinMode (), digitalWrite(), and digitalRead () functions. They operate at 5 volts. Each pin can provide or receive 20 mA as recommended operating condition and has an internal pull-up resistor (disconnected by default) of 20-50k ohm. A maximum of 40mA is the value that must not be exceeded on any I/O pin to avoid permanent damage to the microcontroller.

- In addition, some pins have specialized functions:
  Serial: 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.
- External Interrupts: 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the attach Interrupt () function for details.
- PWM: 3, 5, 6, 9, 10, and 11. Provide 8-bit PWM output with the analogWrite() function.
- SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). These pins support SPI communication using the SPI library.
- LED: 13. There is a built-in LED driven by digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

- TWI: A4 or SDA pin and A5 or SCL pin. Support TWI communication using the Wire library.

The Uno has 6 analog inputs, labeled A0 through A5, each of which provide 10 bits of resolution (i.e. 1024 different values). By default, they measure from ground to 5 volts, though is it possible to change the upper end of their range using the AREF pin and the analog Reference () function.

There are a couple of other pins on the board:

- AREF. Reference voltage for the analog inputs. Used with analogReference().
- Reset. Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

Communication

Arduino/Genuino Uno has a number of facilities for communicating with a computer, another Arduino/Genuino board, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The 16U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, a .inf file is required. The Arduino Software (IDE) includes a serial monitor which allows simple textual data to be sent to and from the board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1).

A Software Serial library allows serial communication on any of the Uno's digital pins.

The ATmega328 also supports I2C (TWI) and SPI communication. The Arduino Software (IDE) includes a Wire library to simplify use of the I2C bus; see the documentation for details. For SPI communication, use the SPI library.

Automatic (Software) Reset

Rather than requiring a physical press of the reset button before an upload, the Arduino/Genuino Uno board is designed in a way that allows it to be reset by software running on a connected computer. One of the hardware flow control lines (DTR) of the ATmega8U2/16U2 is connected to the reset line of the ATmega328 via a 100 nano farad capacitor. When this line is asserted (taken low), the reset line drops long enough to reset the chip. The Arduino Software (IDE) uses this capability to allow you to upload code by simply pressing the upload button in the interface toolbar. This means that the bootloader can have a shorter timeout, as the lowering of DTR can be well-coordinated with the start of the upload.

This setup has other implications. When the Uno is connected to either a computer running Mac OS X or Linux, it resets each time a connection is made to it from software (via USB). For the following half-second or so, the bootloader is running on the Uno. While it is programmed to ignore malformed data (i.e. anything besides an upload of new code), it will intercept the first few bytes of data sent to the board after a connection is opened. If a sketch running on the board receives one-time configuration or other data when it first starts, make sure that the software with which it communicates waits a second after opening the connection and before sending this data.

The Uno board contains a trace that can be cut to disable the auto-reset. The pads on either side of the trace can be soldered together to re-enable it. It's labelled "RESET-EN". You may also be able to disable the auto-reset by connecting a 110 ohm resistor from 5V to the reset line; see this forum thread for details.

Revisions

Revision 3 of the board has the following new features:

- 1.0 pinout: added SDA and SCL pins that are near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage provided from the board. In future, shields will be compatible with both the board that uses the AVR, which operates with 5V and with the Arduino Due that operates with 3.3V. The second one is a not connected pin, that is reserved for future purposes.
- Stronger RESET circuit.
- Atmega 16U2 replace the 8U2.

CHAPTER 3

FUNCTIONAL MODULES & SOFTWARE MODULES DESCRIPTION

3.1 Power Supply:

The input to the circuit is applied from the regulated power supply. The a.c. input i.e., 230V from the mains supply is step down by the transformer to 12V and is fed to a rectifier. The output obtained from the rectifier is a pulsating d.c voltage. So in order to get a pure d.c voltage, the output voltage from the rectifier is fed to a filter to remove any a.c components present even after rectification. Now, this voltage is given to a voltage regulator to obtain a pure constant dc voltage. The block diagram of regulated power supply is shown in the figure 3.2

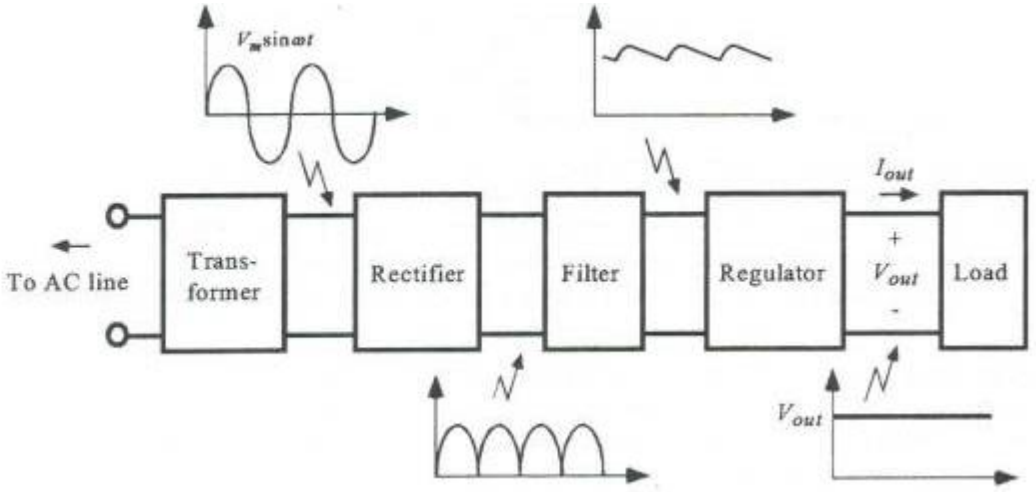

Figure 3.1 components of power supply

Transformer:

Usually, DC voltages are required to operate various electronic equipment and these voltages are 5V, 9V or 12V. But these voltages cannot be obtained directly. Thus, the a.c input available at the mains supply i.e., 230V is to be brought down to the required voltage level. This is done by a transformer. Thus, a step-down transformer is employed to decrease the voltage to a required level.

Rectifier:

The output from the transformer is fed to the rectifier. It converts A.C. into pulsating D.C. The rectifier may be a half wave or a full wave rectifier. In this project, a bridge rectifier is used because of its merits like good stability and full wave rectification.

Filter:

Capacitive filter is used in this project. It removes the ripples from the output of rectifier and smoothens the D.C. Output received from this filter is constant until the mains voltage and load is maintained constant. However, if either of the two is varied, D.C. voltage received at this point changes. Therefore, a regulator is applied at the output stage.

Voltage regulator:

As the name itself implies, it regulates the input applied to it. A voltage regulator is an electrical regulator designed to automatically maintain a constant voltage level. In this project, power supply of 5V and 12V are required. In order to obtain these voltage levels, 7805 and 7812 voltage regulators are to be used. The first number 78 represents positive supply and the numbers 05, 12 represent the required output voltage levels

Accelerometer

An accelerometer is a sensing element that measures acceleration; acceleration is the rate of change of velocity with respect to time. It is a vector that has magnitude and direction. Accelerometers measure in units of g – a g is the acceleration measurement for gravity which is equal to $9.81 m/s^2$. Accelerometers have developed from a simple water tube with an air bubble that showed the direction of the acceleration to an integrated circuit that can be placed on a circuit board. Accelerometers can measure: vibrations, shocks, tilt, impacts and motion of an object.

Types of Accelerometers

There are a number of types of accelerometers. What differentiates the types is the sensing element and the principles of their operation.

Capacitive accelerometers sense a change in electrical capacitance, with respect to acceleration. The accelerometer senses the capacitance change between a static condition and the dynamic state.

Piezoelectric accelerometers use materials such as crystals, which generate electric potential from an applied stress. This is known as the piezoelectric effect. As stress is applied, such as acceleration, an electrical charge is created.

Piezoresistive accelerometers (strain gauge accelerometers) work by measuring the electrical resistance of a material when mechanical stress is applied

Hall Effect accelerometers measure voltage variations stemming from a change in the magnetic field around the accelerometer.

Magneto resistive accelerometers work by measuring changes in resistance due to a magnetic field. The structure and function is similar to a Hall Effect accelerometer except that instead of measuring voltage, the magneto resistive accelerometer measures resistance.

Heat transfer accelerometers measure internal changes in heat transfer due to acceleration. A single heat source is centred in a substrate and suspended across a cavity. Thermo resistors are spaced equally on all four sides of the suspended heat source. Under zero acceleration the heat gradient will be symmetrical. Acceleration in any direction causes the heat gradient to become asymmetrical due to convection heat transfer.

MEMS-Based Accelerometers

MEMS (Micro-Electro Mechanical System) technology is based on a number of tools and methodologies, which are used to form small structures with dimensions in the micrometer scale (one millionth of a meter). This technology is now being utilized to manufacture state of the art MEMS-Based Accelerometers.

Future Accelerometer Advancements

In the next decade, NANO technology will create new applications and dramatically reshape this area of technology.

Applications for Accelerometer

From industry to education, accelerometers have numerous applications. These applications range from triggering airbag deployments to the monitoring of nuclear reactors. There are a number of practical applications for accelerometers; accelerometers are used to measure static acceleration (gravity), tilt of an object, dynamic acceleration, shock to an object, velocity, orientation and the vibration of an object. Accelerometers are becoming more and more ubiquitous: cell phones, computers and washing machines now contain accelerometers.

Other practical applications include:

- Measuring the performance of an automobile

- Measuring the vibration of a machine
- Measuring the motions of a bridge
- Measuring how a package has been handled

Digital accelerometers will give you information using a serial protocol like I2C , SPI or USART, while analog accelerometers will output a voltage level within a predefined range that you have to convert to a digital value using an ADC (analog to digital converter) module.

The digital accelerometer is more sophisticated than the analog. Here the amount of high voltage time is proportional to the acceleration. One of its major advantages is that it is more stable and produces a direct output signal. Accelerometers are now also used in aerospace and many military applications, such as missile launch, weapon fire system, rocket deployment etc. Many a times these accelerometers are used to protect fragile equipment during cargo transportation, and report any strain that might cause a possible damage. Some companies have also managed to develop a wireless 3-axis accelerometers which are not only low in cost but are also shock durable. This 3-axis accelerometer has sensors that are used to protect mobiles and music players. Also these sensors are used in some of the devices used for traffic navigation.

Figure 3.2 ADXL 335

Accelerometer consists of five six pins and we use five pins for our requirement. These five pins are

1)+vcc

2)Gnd

3) x-axis

4) y-axis

5)z-axis

We connect ++vcc to five volts, and another one is for ground. The remaining pins are connected to the adc channels of microcontroller. The output from accelerometer is analog value and we need to convert these values to digital.

By tilting the accelerometer device, we can find the values that are to be changing in accordance with accelerometer tilting.

3.2 BUZZER:

1. When mounting and handling

1) To prevent malfunctions, install the piezoelectric buzzer or sounder so that it does not come into contact with other components on its side or top surface.

(2) Do not block the sound release hole of the piezoelectric buzzer or sounder. Maintain a distance of at least 10mm between the sound release hole and any surrounding object.

Also, do not cover the sound release hole with an adhesive tape or the like. If the sound hole is blocked or covered, the piezoelectric buzzer or sounder may exhibit abnormal oscillation or stop functioning.

(3) The sound pressure of the piezoelectric buzzer or sounder may be measured after, but not before, it is installed in the host equipment.

When determining the installation position, make sure that adverse acoustic impedance does not exist in the installation area. If acoustic impedance exists, the piezoelectric buzzer or sounder may exhibit abnormal oscillation or stop functioning. (4) When screw down the piezoelectric buzzer or sounder, tighten the screws within the specified torque range. Use pan-headed screws and washers not to deform the casing. A deformed casing may cause the piezoelectric buzzer or sounder to exhibit abnormal oscillation or stop functioning.

(5) When stripping a lead wire, do not cut the conductive line inside the coating, thereby ensuring the sound will be properly generated. Use a stripper suitable for

the diameter of the lead wire.

(6) Do not apply strong force to the pins before they are soldered. If the pins are bent or cut due to excessive force, the piezoelectric buzzer or sounder may not generate sound.

(7) Do not connect the piezoelectric buzzer or sounder improperly, otherwise the internal circuit may break down when electricity is applied.

(8) Use the piezoelectric buzzer or sounder within the designated operation voltage range. A higher voltage may damage the diaphragm and other components or cause a fire. With a lower voltage, the sound may not be generated.

(9) Do not apply DC voltage to the piezoelectric sounder. Otherwise, silver migration may occur, which will lower the insulation resistance and cause the sounder to stop functioning.

(10) Use a low-impedance (not more than 100Ω) power supply for the piezoelectric buzzer; otherwise, the piezoelectric buzzer may exhibit abnormal oscillation or stop functioning.

(11) Do not interpose a resistance in series between the piezoelectric buzzer and the power supply, otherwise the piezoelectric buzzer may exhibit abnormal oscillation or stop functioning. If the interposition of a resistance (3kΩ Max.) is necessary to adjust the sound volume, insert a capacitor of approximately 1 μF in parallel, not in series. (Fig.1)

(12) Do not use the piezoelectric buzzer or sounder where any corrosive gas, such as H2S, etc. Otherwise, a normal sound may not be generated due to corrosion of the components and diaphragm.

(13) Do not wash the piezoelectric buzzer or sounder with solvent or allow solvent vapor to enter them while washing. Any solvent trapped inside the casing may damage the piezoelectric buzzer or sounder.

(14) Do not drop the piezoelectric buzzer or sounder. With a mechanical shock, the piezoelectric sounder may accumulate a high voltage inside its piezoelectric elements, resulting in an electric shock to anyone who touches it. Also if such sounder is connected to a circuit, it may damage transistors and/or other electronic components. Sounders which have accidentally gotten a mechanical shock can be made safe by shorting them between the poles. Then check the sound pressure, tone

and appearance before use.

(15) Take special protective measures is required to prevent deterioration and malfunction, whenever the piezoelectric buzzers or sounders are stored in the following unfriendly areas.

①Dusty places

②Hot or frosty places

③Areas exposed to sunlight

④Places with leaking or infiltrating water

⑤Humid places

⑥Areas exposed to solvents or their vapor

(16) When operating the piezoelectric buzzer or sounder outdoors, protect it from moisture to ensure normal operation.

LCD

LCD (Liquid Crystal Display) screen is an electronic display module and find a wide range of applications. A 16x2 LCD display is very basic module and is very commonly used in various devices and circuits. These modules are preferred over seven segments and other multi segment LEDs. The reasons being: LCDs are economical; easily programmable; have no limitation of displaying special & even custom characters (unlike in seven segments), animations and so on.

A **16x2 LCD** means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data. The command register stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD.

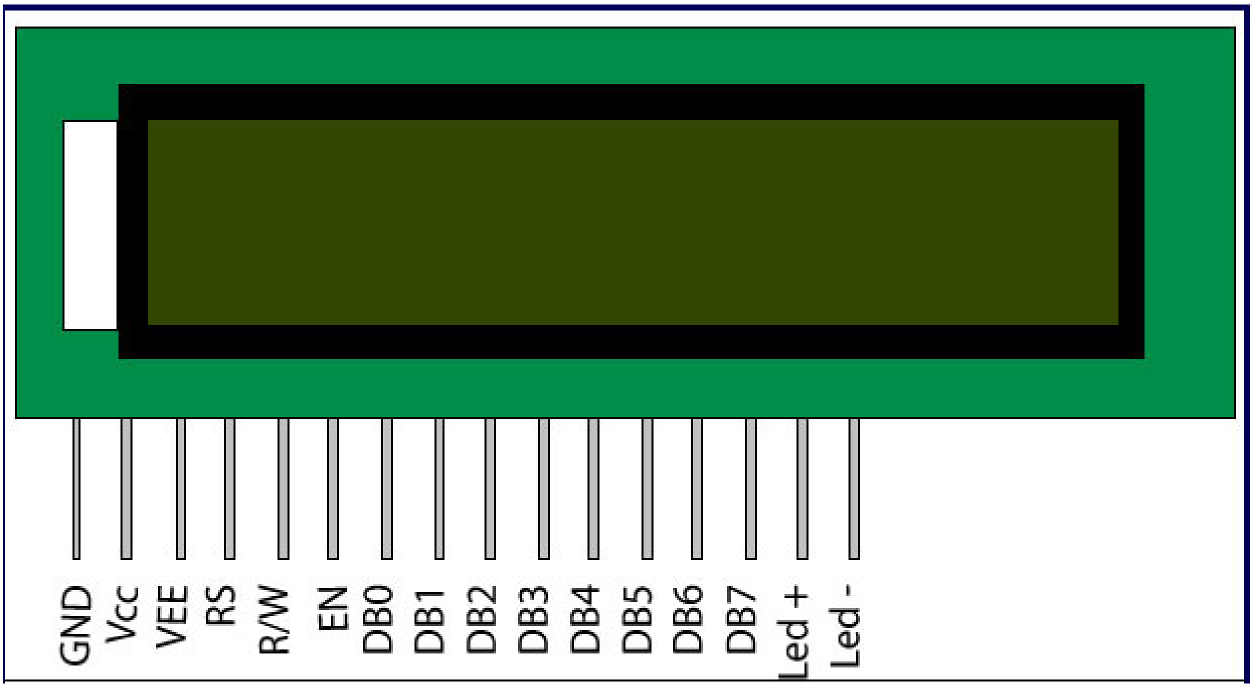

Figure:3.3 16x2 LCD

Pin No	Function	Name
1	Ground (0V)	Ground
2	Supply voltage; 5V (4.7V – 5.3V)	Vcc
3	Contrast adjustment; through a variable resistor	V_{EE}
4	Selects command register when low; and data register when high	Register Select
5	Low to write to the register; High to read from the register	Read/write
6	Sends data to data pins when a high to low pulse is given	Enable
7	8-bit data pins	DB0
8		DB1
9		DB2
10		DB3
11		DB4
12		DB5
13		DB6
14		DB7
15	Backlight V_{CC} (5V)	Led+
16	Backlight Ground (0V)	Led-

Table 3.1 Functions of LCD

3.3 SOFTWARE MODULES DESCRIPTION

3.31 ARDUNIO INSTALLATION:

After learning about the main parts of the Arduino UNO board, we are ready to learn how to set up the Arduino IDE. Once we learn this, we will be ready to upload our program on the Arduino board.

In this section, we will learn in easy steps, how to set up the Arduino IDE on our computer and prepare the board to receive the program via USB cable.

Step 1: First you must have your Arduino board (you can choose your favorite board) and aUSB cable. In case you use Arduino UNO, ArduinoDuemilanove, Nano, Arduino Mega 2560, or Diecimila, you will need a standard USB cable (A plug to B plug), the kind you would connect to a USB printer as shown in the following image.

Step 2: Download Arduino IDE Software.

You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with

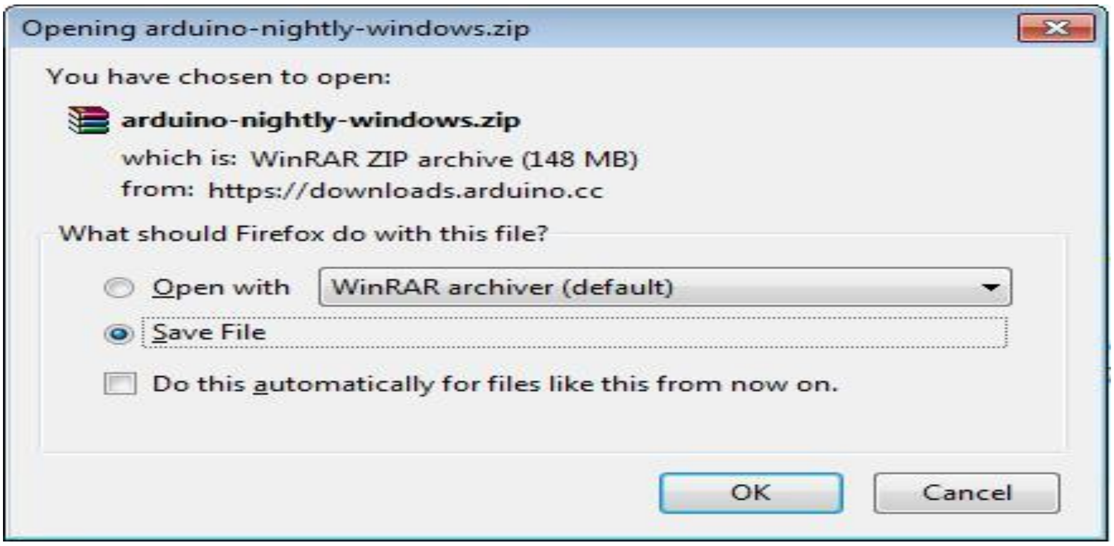

Figure:3.4 Arduino zip

your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.

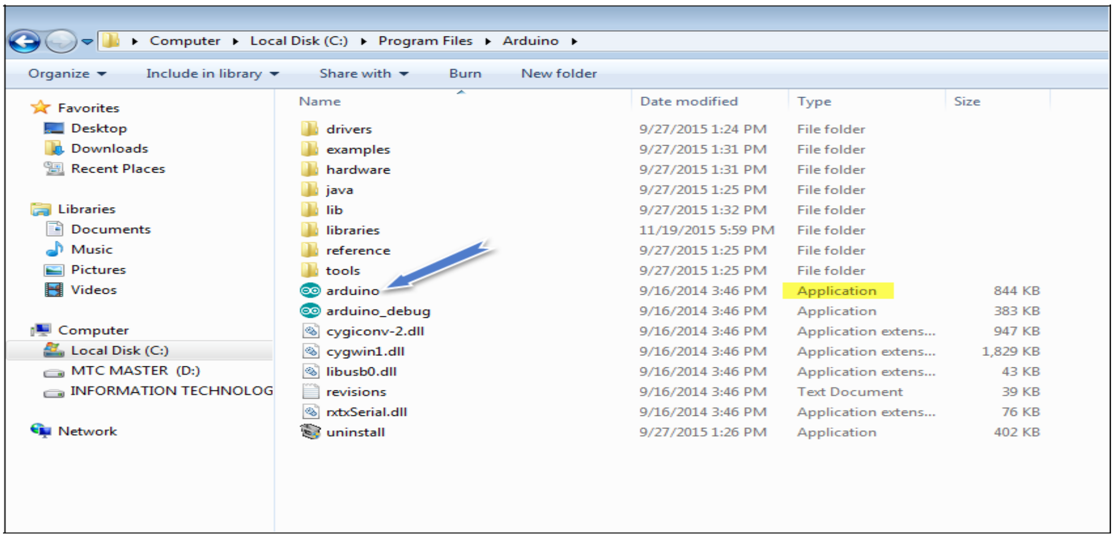

Figure:3.5 Arduino file in windows

Step 3: Power up your board. The Arduino Uno, Mega, Duemilanove and Arduino Nano automatically draw power from either, the USB connection to the computer or an external power supply. If you are using an Arduino Diecimila, you have to make sure that the board is configured to draw power from the USB connection. The power source is selected with a jumper, a small piece of plastic that fits onto two of the three pins between the USB and power jacks. Check that it is on the two pins closest to the USB port.

Connect the Arduino board to your computer using the USB cable. The green power LED (labeled PWR) should glow.

Step 4: Launch Arduino IDE. After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Double-click the icon to start the IDE.

Step 5: Open your first project.

Once the software starts, you have two options:

Create a new project.

Open an existing project example.

To create a new project, select File --> New

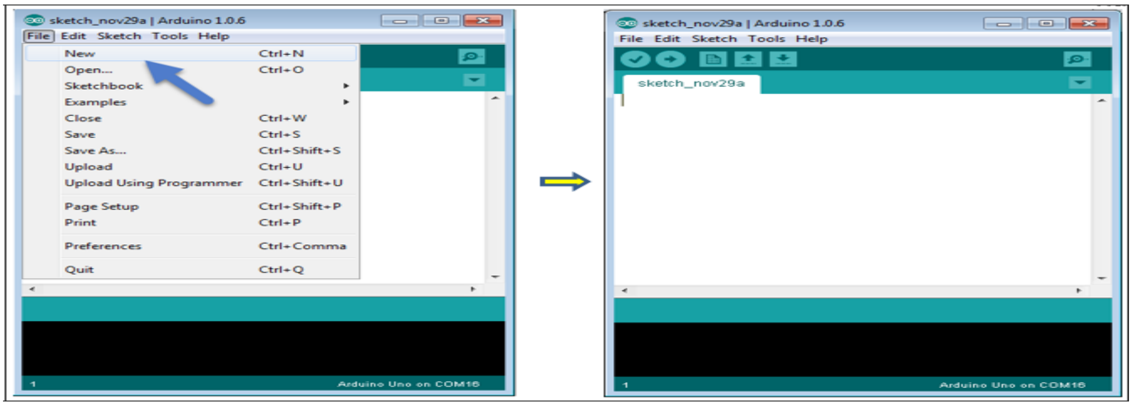

Figure:3.6 Creating a new file

To open an existing project example, select File -> Example -> Basics -> Blink.

Here, we are selecting just one of the examples with the name **Blink**. It turns the LED on and off with some time delay. You can select any other example from the list.

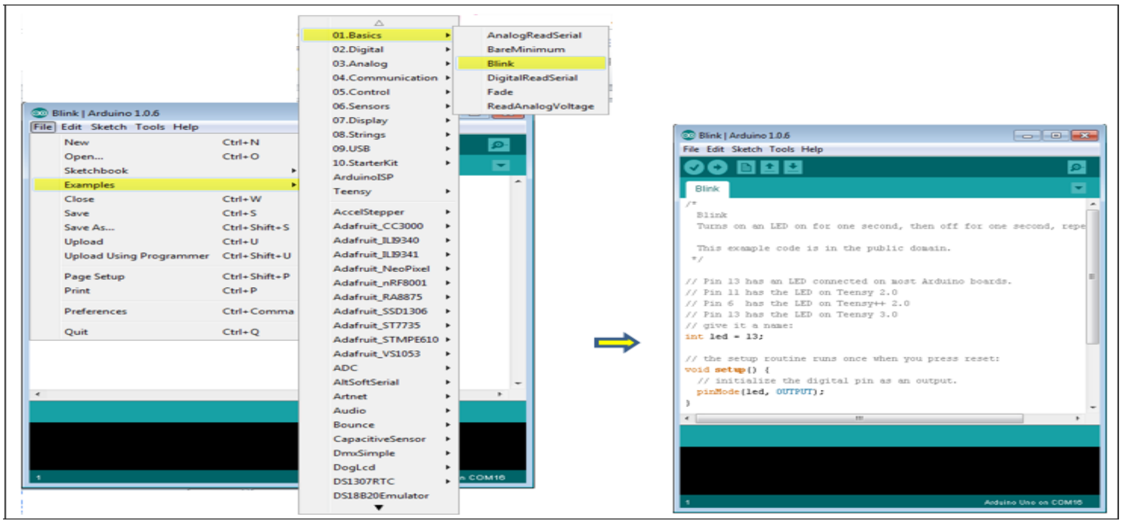

Figure 3.7 Creating File Blink

Step 6: Select your Arduino board.

To avoid any error while uploading your program to the board, you must select the correct Arduino board name, which matches with the board connected to your computer.

Go to Tools -> Board and select your board

CHAPTER 4
PROJECT IMPLEMENTATION

4.1 Design and implementation:

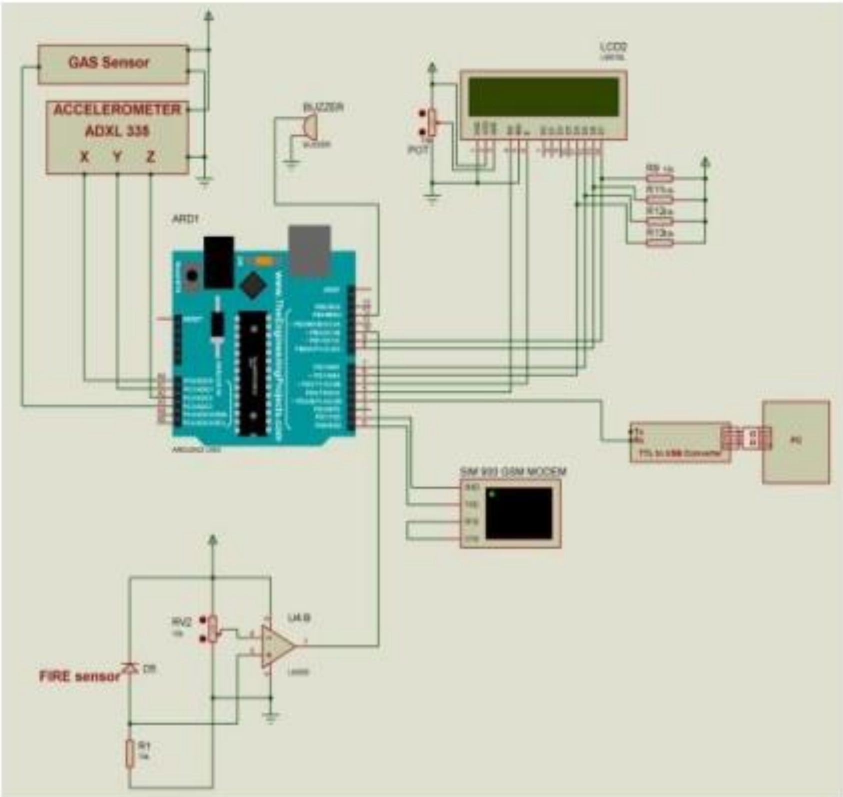

Figure 4.1: Circuit Diagram of the project

Use an accelerometer to determine the significant vibrations of an earthquake along one of the three axes. Therefore, any vibration that occurs is detected by the accelerometer and converted to an equivalent ADC value [1]. These values are read by the Arduino and displayed on the 16x2 LCD. Accelerometer calibration is done by taking environmental vibration samples when the Arduino is turned on. This formula then needs to be subtracted from the actual reading to get the actual reading. Therefore, continuous vibration does not leave any warning around it [7]. After finding the correct reading, the Arduino compares these values with the predefined minimum and maximum values. If the Arduino detects a change for an axis in both directions (negative and positive) greater or less than the specified value, the Arduino will activate the buzzer and show the alarm status on the 16x2 LCD and the LED will also turn on. The sensitivity of the earthquake detector can be adjusted by changing the predefined value in the Arduino code [11]. Flame sensors are used to detect and respond to fire or flame formation. A gas sensor is a device that interacts with gas to measure its concentration and detect it in an area.

4.2 ADVANTAGES

1) Low design time.

2) Low production cost.

3) This system is applicable for both the indoor and outdoor environment.

4) Setting the destination is very easy.

5) It is dynamic system.

6) Less space.

7) Low power consumption.

CHAPTER 5
CONCLUSION AND FUTURE SCOPE

5.1Conclusion:

So, to summarize, the purpose of this product is to reduce the damage caused by earthquakes by warning people. It is very economical and can accommodate every people. This device is designed as a modern technology to solve the problem of automatic detection and classification of earthquakes in one step by using Arduino as an earthquake detector. In this process, most of them do well in protecting lives and resources during earthquakes. The device is small and light, which makes it easy to carry and place anywhere. The, hardware used in this device is very good and uncomplicated. In addition, the system's sensors are very sensitive, allowing for more accurate measurements. The equipment is more efficient and therefore uses less energy. Fire sensor and gas sensor are also connected to it to detect gas and fire environment, so this system has many uses. As a result, the system is very efficient.

5.2 FUTURE SCOPE

. Future scope includes improving the accuracy of the system, and enhancing the mobility of device so that the system can be connected to a more convenient location in a home. We can expand the scale of the project by making collaborations with disaster management agencies.

CHAPTER 6
RESULT & BIBLIOGRAPHY

6.1 RESULT:

An efficient system constructed which would be able to detect the pre-earth quake vibrations using accelerometer ADXL335 sensor; by using the micro controller Arduino we can assess the values of the accelerometer. The equipment is observed to be using lesser energy.

6.2 BIBLIOGRAPHY:

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